# ml

Tanvir

(data + learning algorithm) $\rightarrow$ function

# 1 Supervised learning

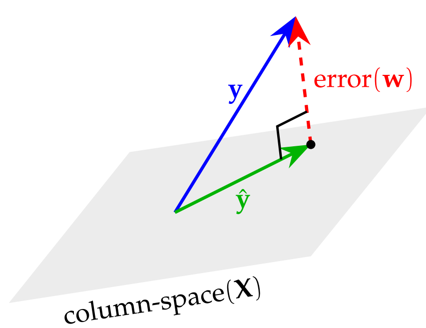
## 1.1 Regression

model, parameters, cost function, objective

### 1.1.1 Linear (in w) Regression

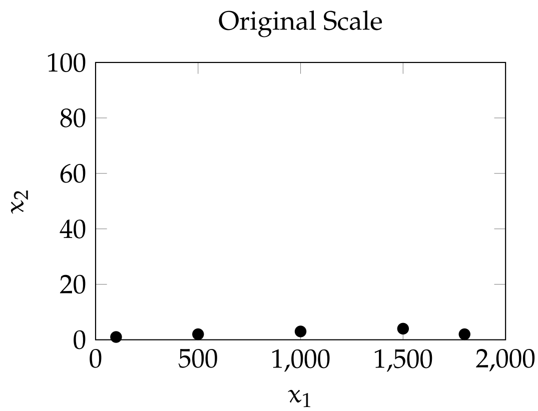
$\mathbf{X}^{m \times n}$ has $m$ examples, each having $n$ features. Usually $m \gg n$. $\mathbf{y}^{m \times 1}$ are the corresponding outputs.

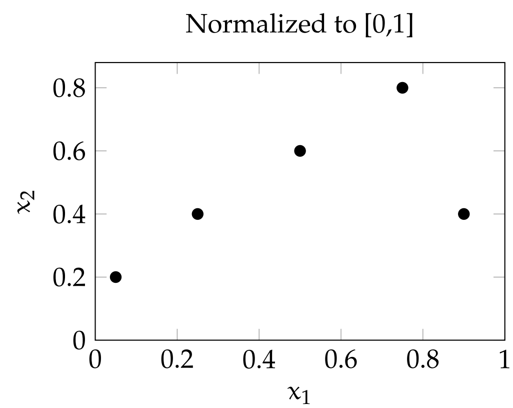
$$\begin{aligned}
\mathbf{X}\mathbf{w} &= \hat{\mathbf{y}} \\
\text{error}(\mathbf{w}) &= \mathbf{y} - \hat{\mathbf{y}} \\
\underset{\mathbf{w}}{\texttt{minimize}} \;\; J(\mathbf{w}) &= \frac{1}{m}\|\text{error}(\mathbf{w})\|^2 = \frac{1}{m}(\mathbf{X}\mathbf{w} - \mathbf{y})^\mathsf{T}(\mathbf{X}\mathbf{w} - \mathbf{y}) \\
\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} &= \frac{2}{m}\mathbf{X}^\mathsf{T}(\mathbf{X}\mathbf{w} - \mathbf{y}) \\
\texttt{gradient descent:} \quad \mathbf{w} &= \mathbf{w} - \alpha \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}
\end{aligned}$$

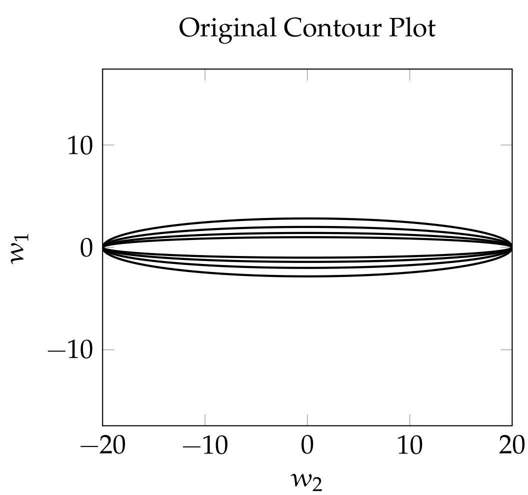


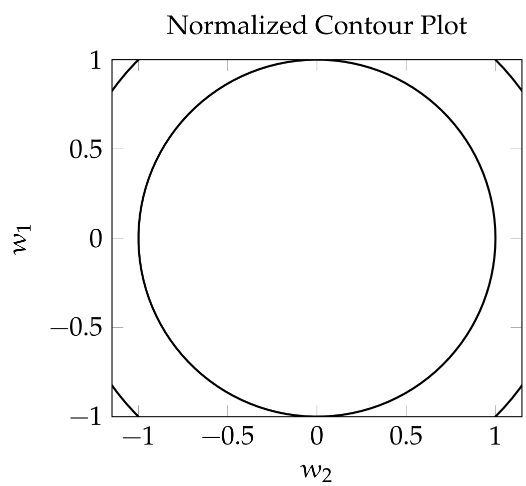
## 1.2 Normalization

Changes scale keeping the shape of distribution same. Gradient descent now works, with more ease, in the world of concentric circular contours.



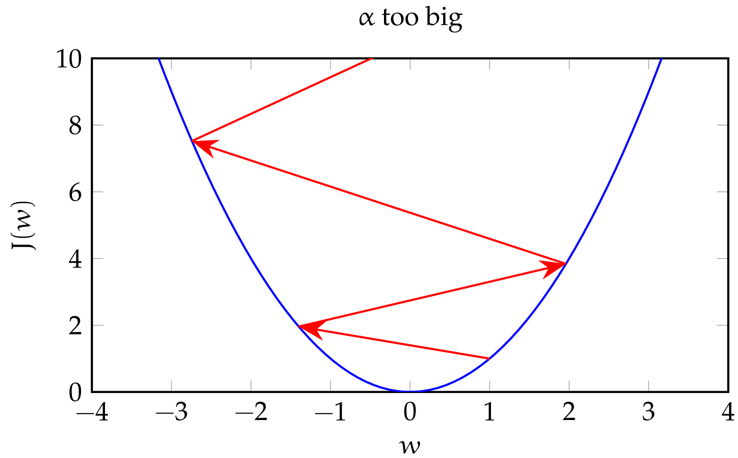



## 1.3 Learning rate, $\alpha$

A good value is somewhere between too big (divergence) and too small (slow convergence).

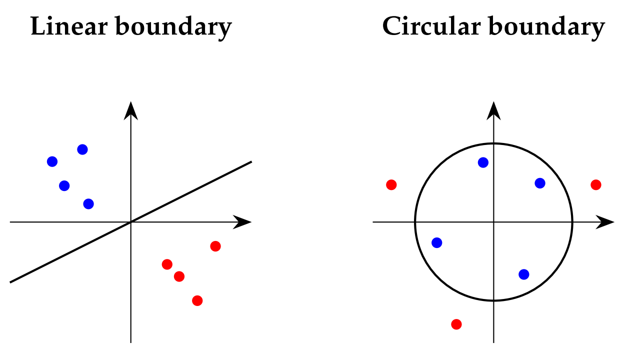



## 1.4 Classification

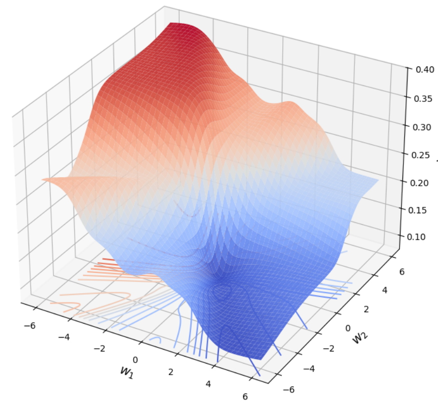
### 1.4.1 Logistic Regression

$$\begin{aligned} \mathbf{z} &= \mathbf{X}\mathbf{w}, && \texttt{// linear regression} \\ \sigma(\mathbf{z}) &= \frac{1}{1+e^{-\mathbf{z}}} && \texttt{// maps z to probability-like (0,1)} \end{aligned}$$

At $\mathbf{z} = \mathbf{0}$, $\sigma(\mathbf{z}) = 0.5$. So, with threshold 0.5, $\mathbf{z} = \mathbf{0}$ is the decision boundary. As linear regression doesn't have to be straight line, logistic regression's decision boundary does not have to be a straight line.



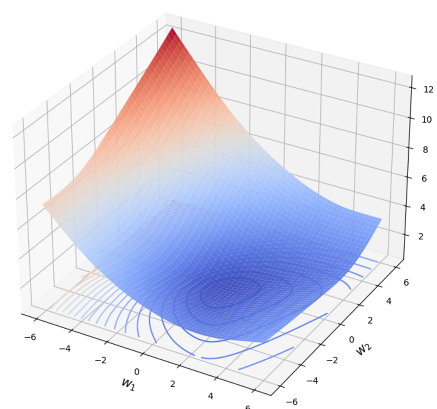
The squared error loss function $L(\mathbf{w}, y_i) = \left(\sigma(\mathbf{w}^T \mathbf{x}_i) - y_i\right)^2$, makes the cost function non-convex.



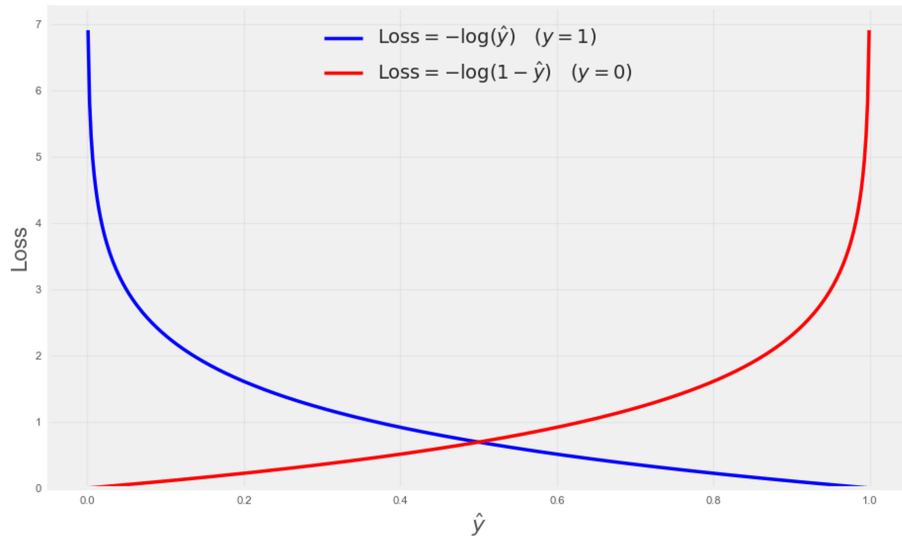

Negative log likelihood loss:

$$\hat{y}_i = \sigma\left(\mathbf{w}^T \mathbf{x}_i\right) \in (0, 1)$$
$$J(\mathbf{w}) = \frac{1}{m} \sum_{i=1}^{m} \begin{cases} -\log(\hat{y}_i), & \text{if } y_i = 1, \\ -\log(1 - \hat{y}_i), & \text{if } y_i = 0 \end{cases}$$

The negative log likelihood loss function with $\sigma(z)$ makes the cost function convex.



Negative log likelihood also incentivizes appropriately: big mismatch $\implies$ big loss, small mismatch $\implies$ small loss.




$$z^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)}$$

Loss per example, $L\left(\mathbf{x}^{(i)}, \mathbf{w}\right) = -y^{(i)} \log\left(\sigma\left(z^{(i)}\right)\right) - \left(1 - y^{(i)}\right) \log\left(1 - \sigma\left(z^{(i)}\right)\right)$

$$\frac{\partial \sigma\left(z^{(i)}\right)}{\partial \mathbf{w}} = \frac{\partial \sigma(z^{(i)})}{\partial z^{(i)}} \cdot \frac{\partial z^{(i)}}{\partial \mathbf{w}} = \sigma(z^{(i)})(1 - \sigma(z^{(i)}))\mathbf{x}^{(i)}$$

$$\frac{\partial L\left(\mathbf{x}^{(i)}, \mathbf{w}\right)}{\partial \mathbf{w}} = \left(\sigma\left(z^{(i)}\right) - y^{(i)}\right)\mathbf{x}^{(i)}$$

$$\frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m} \sum_{i=1}^{m} \frac{\partial L\left(\mathbf{x}^{(i)}, \mathbf{w}\right)}{\partial \mathbf{w}}$$

More concisely,

$$\mathbf{z} = \mathbf{X}\mathbf{w}$$

$$J(\mathbf{X}, \mathbf{w}) = -\frac{1}{m}\left[\mathbf{y}^T \log(\sigma(\mathbf{z})) + (\mathbf{1} - \mathbf{y})^T \log(1 - \sigma(\mathbf{z}))\right]$$

$$\frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m}\mathbf{X}^T(\sigma(\mathbf{z}) - \mathbf{y})$$

Gradient descent: $\mathbf{w} = \mathbf{w} - \alpha \frac{\partial J(\mathbf{X}, \mathbf{w})}{\partial \mathbf{w}}$

### 1.4.2 Error metrics

| | | Predicted: Positive | Predicted: Negative |
|---|---|---|---|
| Actual | Positive | TP | FN |
| | Negative | FP | TN |

**Metrics**

Accuracy: $\dfrac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}$

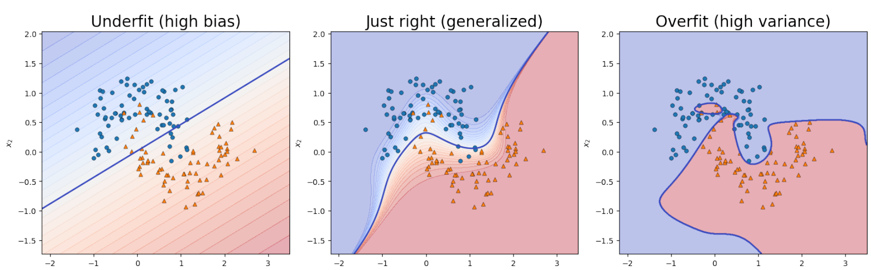
Precision: $\dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$

Recall: $\dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$

F1 score: $\dfrac{2\cdot\mathrm{TP}}{2\cdot\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}$

- Accuracy: How many predictions are correct. Doesn't work when classes are highly imbalanced.
- Precision: $\Pr(\text{positive example} \mid \text{positive prediction})$
- Recall: $\Pr(\text{positive prediction} \mid \text{positive example})$
- F1 score: Harmonic mean of precision and recall. Cannot be high if either precision or recall is low.

# 2 Generalization




## 2.1 Address overfit

1. Collect more training examples keeping the model as it is.
2. Select a smaller set of informative features to simplify the model. Regularizarion disables features in a gradual manner.

### 2.1.1 Regularization

**L2:**

$$
\begin{aligned}
\underset{\mathbf{w}}{\texttt{minimize}}\ \ J(\mathbf{w}) &= \frac{1}{m}\|\text{error}(\mathbf{w})\|^2 \\
\texttt{gradient descent:}\quad \mathbf{w} &= \mathbf{w} - \alpha\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} \\
\underset{\mathbf{w}}{\texttt{minimize}}\ \ J_{\text{reg}}(\mathbf{w}) &= \frac{1}{m}\left(\|\text{error}(\mathbf{w})\|^2 + \lambda\|\mathbf{w}\|^2\right) \\
\texttt{gradient descent (reg):}\quad \mathbf{w} &= \underbrace{\left(1 - \alpha\cdot\frac{2\lambda}{m}\right)}_{\text{shrinkage}}\mathbf{w} - \alpha\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} \qquad \texttt{//}\ \ (\alpha\cdot{}^{2\lambda}/_{m}) \sim 0
\end{aligned}
$$

Here, $\lambda\|\mathbf{w}\|^2$ encourages smaller parameter values. With bigger $\lambda$'s, the parameters approach zero.

We can think regularization in terms of a constrained optimization problem:

$$
\begin{aligned}
\underset{\mathbf{w}}{\texttt{minimize}}\ \ & J(\mathbf{w}) \\
\texttt{with constraint:}\quad & \|\mathbf{w}\|^2 \leqslant r^2
\end{aligned}
$$

In other words, minimize $J(\mathbf{w})$ but do not let $\mathbf{w}$ go beyond a ball of radius $r$.

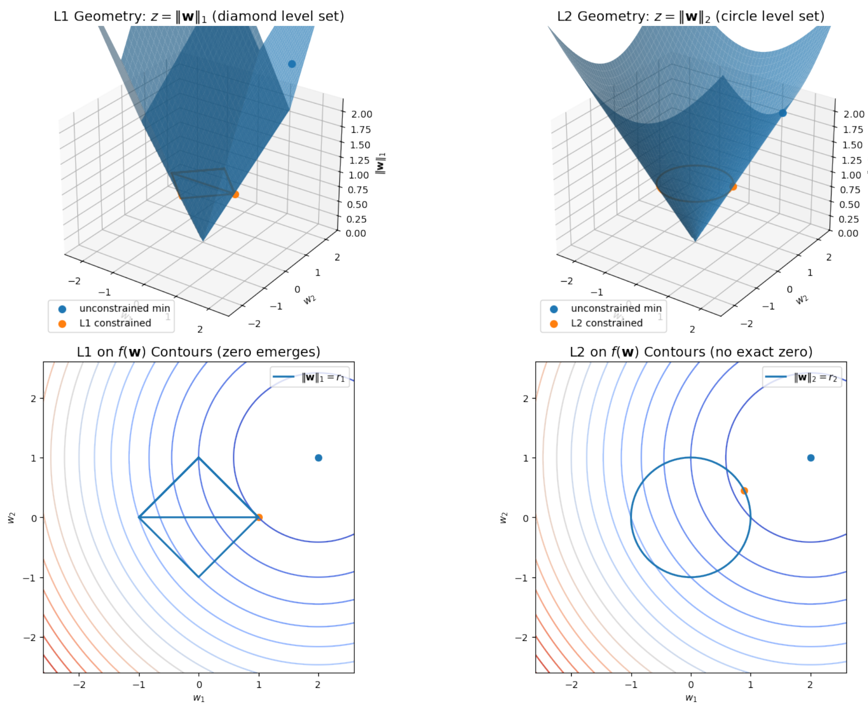
**L1:**

$$
\begin{aligned}
\underset{\mathbf{w}}{\texttt{minimize}}\ \ J_{\text{reg}}(\mathbf{w}) &= \frac{1}{m}\left(\|\text{error}(\mathbf{w})\|^2 + \lambda\|\mathbf{w}\|_1\right) \\
\|\mathbf{w}\|_1 &= \sum_i |w_i| \\
\texttt{gradient descent (reg):}\quad \mathbf{w} &= \mathbf{w} - \underbrace{\alpha\cdot\frac{\lambda}{m}\,\text{sign}(\mathbf{w})}_{\substack{\text{constant pull}\\ \text{towards 0}}} - \alpha\cdot\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}
\end{aligned}
$$

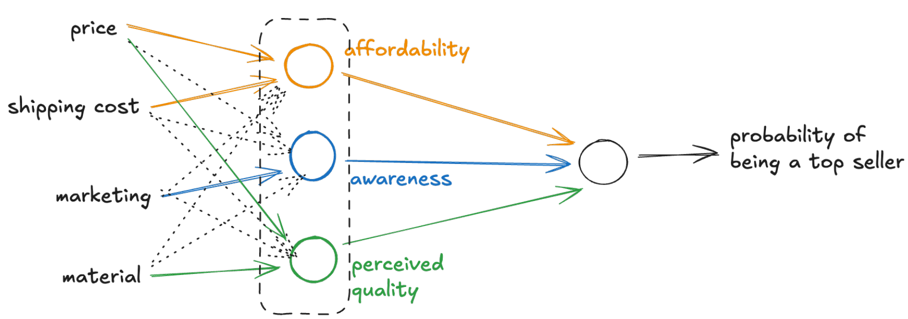
The corresponding constrained optimization:

$$
\begin{aligned}
\underset{\mathbf{w}}{\texttt{minimize}}\ \ & J(\mathbf{w}) \\
\texttt{with constraint:}\quad & \|\mathbf{w}\|_1 \leqslant r
\end{aligned}
$$

With L1 regularization, parameters for less important features can be exactly zero.




# 3 Neural Networks




- A neural network realizes a piecewise differentiable function.
- In multi-layer perceptron (MLP), an individual neuron can do linear or logistic regression.
- Intermediate layers of neurons pick out features in a hierarchical manner. Left to right, the features become more abstract – feature from inputs, feature from features.

- Unlike other ML algorithms, a neural network's performance scales well with data.

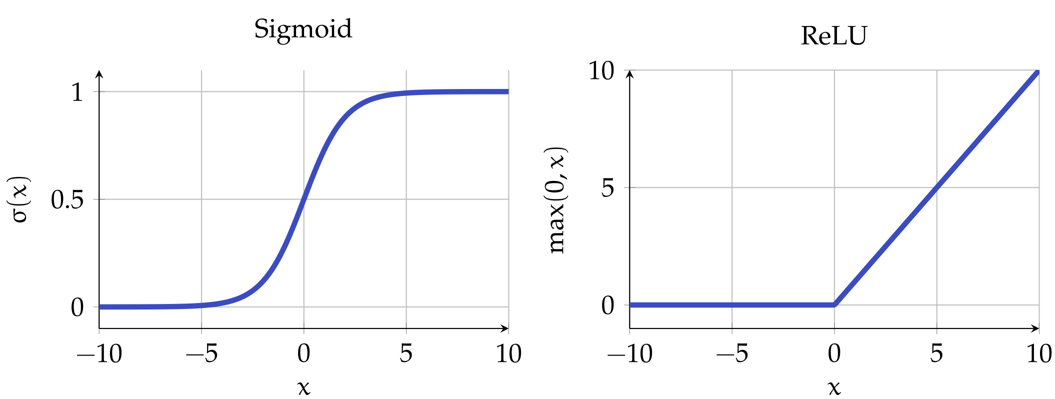
```
layer_1 = keras.layers.Dense(
    units=3,
    activation="sigmoid"
)
layer_2 = keras.layers.Dense(
    units=1,
    activation="sigmoid"
)
model = keras.Sequential([layer_1, layer_2])
model.compile(optimizer="adam", loss="mse")
model.fit(X, y, epochs=100)
model.predict(X_new)
```

## 3.1 Loss function

Target of optimization. For binary classification, we can use "binary cross entropy". For regression, we can use "mean squared error".

## 3.2 Activation function

Without activation function, a neural network can only implement a linear function. For example, it cannot implement the `XOR` function.




- Sigmoid$(x) \in (0, 1)$ vs. ReLU$(x) \in [0, \infty)$. So, ReLU$(x)$ allows large positive outputs.
- Output layer:
  - Binary classification: Sigmoid
  - Non-negative Regression: ReLU

  - Regression: None.
- Hidden layer: ReLU. Because:
  - Faster to compute
  - Flat regions have near-zero gradients, so gradient descent takes tiny steps and progresses slowly there. Sigmoid has two flat regions, ReLU has one.
- The off feature (0-region) of ReLU enables a neural network to stitch together linear segments to model a complex non-linear function.

#### 3.2.1 Softmax

For multi-class classification. For $n$ classes, activation of one of the $n$ output neurons:

$$z_j = \mathbf{w}_j \cdot \mathbf{x} + b_j,\ j = 1, 2, \ldots, n \quad \texttt{// logits}$$

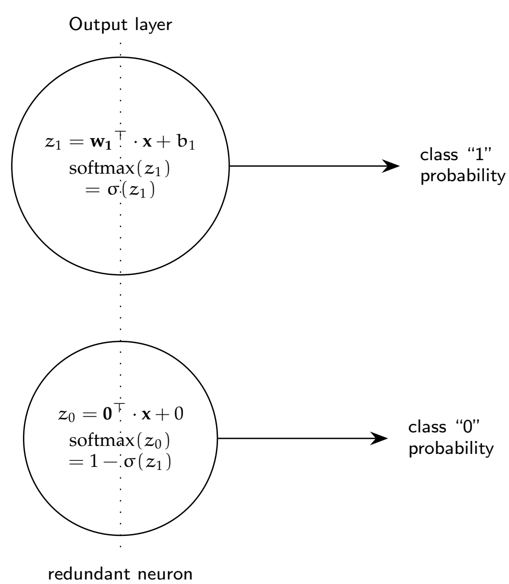
$$a_j = \Pr(y = j \mid \mathbf{x}) = \mathrm{Softmax}(z_j) = \frac{e^{z_j}}{\sum_{k=1}^{n} e^{z_k}}$$

- $\mathrm{Softmax}(z + c) = \mathrm{Softmax}(z)$
- When $n = 2$, Softmax $\equiv$ Sigmoid:

$$a_1 = \mathrm{Softmax}(z_1) = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} = \frac{1}{1 + e^{-(z_1 - z_0)}}$$

When $z = 0$, $a_1 = \mathrm{Sigmoid}(z_1), a_0 = 1 - a_1$

As if we have two output neurons, one with all parameters set to 0.



Training logits and applying Softmax at inference time improves numerical stability:

```
model = Sequential(
    [
        Dense(units=25, activation="relu"),
        Dense(units=15, activation="relu"),
        # outputs logits
        Dense(units=10, activation="linear")
    ]
)
model.compile(
    optimizer="adam",
    # train logits (z's)
    loss=SparseCategoricalCrossentropy(from_logits=True)
)
mode.fit(X, y, epochs=100)
logits = model(X_new)
probabilities = tf.nn.softmax(logits)
```

## 3.3 Optimizer

### 3.3.1 Gradient descent

$$\mathbf{w} = \mathbf{w} - \alpha\,\boldsymbol{\nabla} J(\mathbf{w})$$

Learning rate $\alpha$ is fixed.

### 3.3.2 Adam (ADAptive Moment estimation)

$$\mathbf{w} = \mathbf{w} - \mathbf{A}\,\boldsymbol{\nabla} J(\mathbf{w}), \text{ with } \mathbf{A} = \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_n)$$

- Generally faster convergence than gradient descent.
- Each parameter has its own learning rate.
- Each learning rate adapts on the fly based on earlier steps.

## 3.4 Layers

### 3.4.1 Dense layer

Activation of a neuron is a function of the activations of *all neurons* in the previous layer.

### 3.4.2 Convolutional layer

Activation of a neuron is a function of the activations of *a limited number of neurons* (overlapping sliding window) in the previous layer.

- Faster computation
- Less prone to overfitting

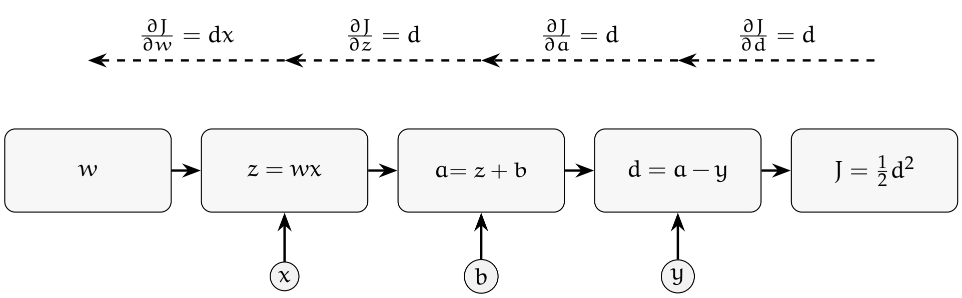
## 3.5 Backpropagation

```
J, w = sympy.symbols("J, w")
J = 1 / (1 + sympy.exp(-w))
# symbolic differentiation
dJ_dw = sympy.diff(J, w)
dJ_dw.subs([(w, 2)])
```

### 3.5.1 Computation graph

Evaluate DAG in topological order in forward-prop and in topological order of the reversed DAG in backprop.

$$
\begin{aligned}
J &= \frac{1}{2}d^2 \\
d &= a - y \\
a &= z + b \\
z &= wx
\end{aligned}
$$

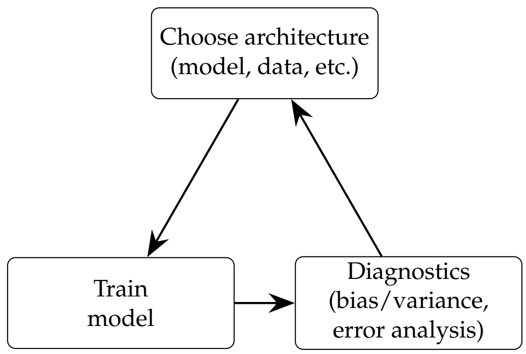



With $N$ nodes and $P$ parameters, $T(\text{backprop}) = \mathcal{O}(N + P)$

### 3.5.2 Autodiff

Tensorflow (or PyTorch) takes care of backprop for us via automatic differentiation.

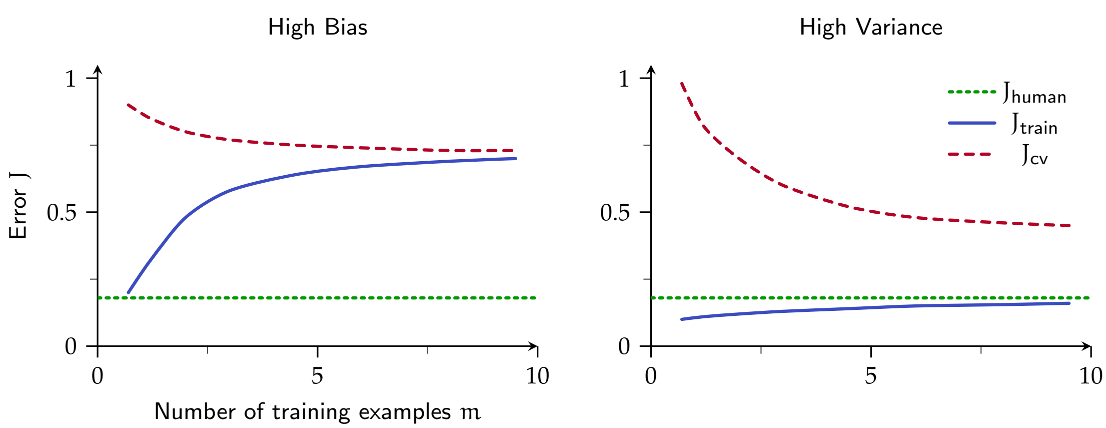
# 4 Debugging learning algorithms




## 4.1 Bias and variance

- High bias: $J_{\text{train}}$ is large. $J_{\text{train}} \approx J_{\text{cv}}$. More data alone isn't going to help.
- High variance: $J_{\text{train}}$ is small. $J_{\text{cv}} \gg J_{\text{train}}$. More data alone can help.
- High bias and high variance: $J_{\text{train}}$ is large. $J_{\text{cv}} \gg J_{\text{train}}$. For part of the input high bias, for some other part of the input high variance. Can happen with neural networks.
- With regularization: too big $\lambda$ causes high bias, too small $\lambda$ causes high variance.
- Large $(J_{\text{train}} - J_{\text{human}}) \implies$ high bias. Large $(J_{\text{train}} - J_{\text{cv}}) \implies$ high variance.

### 4.1.1 Learning curve



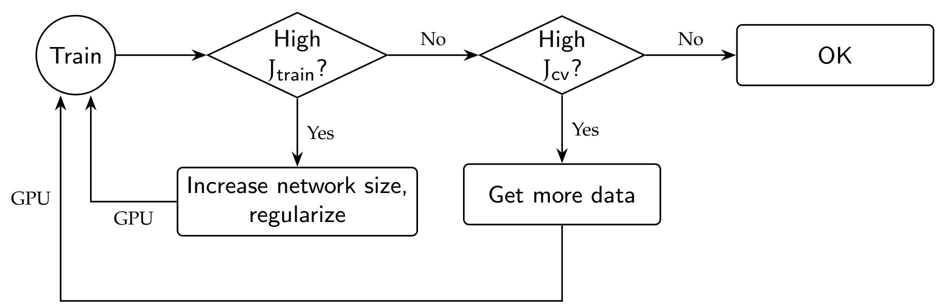
### 4.1.2 Regularized linear regression

$$J(\mathbf{w}) = \frac{1}{m}(\mathbf{Xw} - \mathbf{y})^T(\mathbf{Xw} - \mathbf{y}) + \lambda\frac{1}{m}\|\mathbf{w}\|^2$$

Address large error in predictions:

| High bias | High variance |
|---|---|
| Try getting a bigger set of features | Try getting a smaller set of features |
| Try adding polynomial features | Try getting more training examples |
| Try decreasing $\lambda$ | Try increasing $\lambda$ |

### 4.1.3 Neural networks (NN)




A large NN has low bias. So, a large NN with regularization may work well:

$$J(\mathbf{W}) = \frac{1}{m}\sum_{i=1}^{m} \text{loss}_i + \lambda\frac{1}{m}\sum_{\mathbf{w}\in\mathbf{W}} \|\mathbf{w}\|^2$$

```
layer = Dense(
    units=25,
    activation="relu",
    # λ = 0.01
    kernel_regularizer=L2(0.01)
)
```

## 4.2 Error analysis

If the learning algorithm misclassified 100 examples (sample of 100 if too many errors) in the validation set, look deeply at these examples and categorize them based on common traits. Address bigger categories first.

### 4.3 Adding data

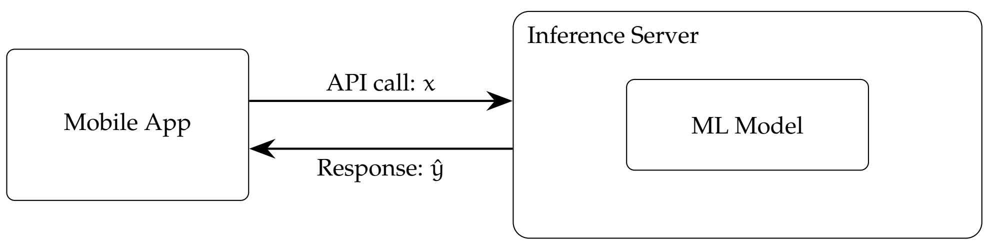
$$\mathsf{ML} = \mathsf{Code} + \mathsf{Data}$$

- Collect more data for the bigger categories from error analysis.
- Data augmentation: keep label same.
  - Rotate, distort, or re-color images.
  - Add crowd noise sound and voice snippet to make noisy speech.
- Synthetic data.

## 5 Transfer learning

Use weights of the initial layers (generic features) of an already trained (for a similar task) neural network (*pretraining*) to either initialize weights or as frozen, trained layers for the current task (*fine tuning*).

## 6 Deployment




## 7 Fairness, bias, ethics

## 8 References

- Machine learning specialization – Coursera